

AHOURET DONAMBI THIERRY

Data Engineer

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PROFILE

Data Engineer and AWS Certified Solutions Architect completing an MSc in Big Data Analytics, with hands-on experience building batch and real-time pipelines on Kafka, PySpark and HDFS, and modelling time-series and transactional data in PostgreSQL/TimescaleDB, MySQL and AWS RDS. Comfortable owning a pipeline end to end ingestion, transformation, storage, containerised orchestration, and the dashboard on top of it. Bilingual French/English, seeking Data Engineering roles building scalable, reliable data systems.

TECHNICAL SKILLS

Languages Python, SQL, R, JavaScript, Bash

Big Data & Streaming Apache Spark (PySpark, Structured Streaming, MLlib), Apache Kafka (KRaft), Hadoop HDFS

Databases PostgreSQL, TimescaleDB, MySQL, AWS RDS, SQLAlchemy, schema design & query optimisation

Cloud & Tooling AWS (EC2, S3, RDS), Docker & Docker Compose, Linux, Git, Jupyter, Postman

Data & Analytics Pandas, NumPy, ETL/ELT design, Grafana, Matplotlib, Seaborn, Chart.js, openpyxl

Backend & APIs Django, Django REST Framework, RESTful API design, htmx, Bootstrap, Tailwind

PROFESSIONAL EXPERIENCE

Freelance Data & Full-Stack Engineer

Dec 2025 – Present

Freelance (Self-Employed)

Kigali, Rwanda

- Own end-to-end delivery for commercial clients, NGOs and data initiatives requirements, data modelling, backend architecture, deployment and handover.
- Designed and shipped **VetConnect Rwanda**, a 10-app Django platform linking livestock farmers to licensed veterinarians: UUID primary keys, phone-number authentication, subscription gating, mobile-money webhook ingestion and full EN/FR/RW localisation via gettext.
- Built **Mr. Cocorico** (mrcocorico.com), a butchery management and POS system on an immutable StockMovement ledger with weighted-average-cost COGS computed in a dedicated services layer auditable stock history rather than mutable balances.
- Architected **PeaceData Portal**, a peacebuilding data platform with role-based access control across project managers, field data engineers and analysts, plus a five-chart analytics dashboard and an Excel export pipeline (openpyxl).
- Manage backend database architecture across PostgreSQL and MySQL, tune SQL queries and integrate REST APIs in production Django environments.

Network Operations Center Technician (L1)

Jan 2025 – Jun 2025

ZTE Rwanda Limited

Kigali, Rwanda

- Monitored network alarms and KPIs across live infrastructure, triaging and prioritising faults to minimise downtime and service disruption.
- Produced daily network health reports by aggregating performance metrics into actionable summaries for wireless product teams and management.
- Coordinated with site engineers on hardware anomalies and escalated complex incidents, improving response times.

IT Technician Intern

Aug 2022 – Oct 2022

SOMA Foundation Rwanda

Kigali, Rwanda

- Configured and deployed portable routing hardware, applied firmware updates and resolved connectivity issues supporting daily operations.

DATA ENGINEERING PROJECTS

Real-Time eCommerce Purchase Prediction Pipeline | *Kafka, PySpark Structured Streaming, HDFS, Spark MLlib, MySQL, Django, Docker Compose*

- Built a fully containerised end-to-end pipeline (HDFS namenode/datanode, Kafka in KRaft mode, ingestion, streaming and training jobs, MySQL, Django dashboard) replaying a month of clickstream events as a live stream.
- Trained a class-weighted logistic regression on **9.4M sessions** (6.7% purchase rate), reaching **0.828 test AUC** at a tuned decision threshold.

- Implemented a Champion/Challenger retraining loop writing model metadata to a MySQL model_registry table, with the streaming job hot-reloading the active model without redeployment.

Energy Consumption Monitoring Platform | PostgreSQL 17, TimescaleDB, Grafana, Python

- Modelled roughly **8M readings from 1,000 meters** as TimescaleDB hypertables, with timezone-aware (Africa/Kigali) time-bucket aggregations and region grouping derived from meter identifiers.
- Benchmarked query performance across candidate front-ends (Streamlit, Dash, Grafana) and standardised on Grafana with a least-privilege read-only database role.

Automated Data Cleaning Pipeline | Python, Pandas, SQLAlchemy, PostgreSQL

- Ingested and cleaned a 10,000-row sales dataset missing-value handling, type coercion and deduplication cutting data inconsistencies by 7.75% before loading into PostgreSQL via SQLAlchemy.

Smart Greenhouse IoT Monitoring System | Django REST Framework, AWS RDS (MySQL), Chart.js

- Architected a REST ingestion API for multi-zone sensor readings persisted to AWS RDS, with a real-time dashboard and automated SMTP alerting on temperature and humidity threshold breaches.

UNMAPPED Informal Skills Matching | Python, LLM APIs, Streamlit World Bank Youth Summit × Hack-Nation

- Built a three-stage inference pipeline (Skills Signal Engine, AI Readiness Lens, Opportunity Matcher) with JSON-schema-constrained outputs grounded in ILOSTAT and World Bank data.
- Parallelised modules with ThreadPoolExecutor and added hash-based cache invalidation to cut latency and repeat API cost.

EDUCATION

MSc, Big Data Analytics

Dec 2024 – Dec 2026

Adventist University of Central Africa

Kigali, Rwanda

- Coursework: Big Data Essentials, Machine Learning, Database Systems, Data Mining, Cloud Computing & Virtualisation, Statistical Analysis, Web Services & API Development.

BSc, Information Technology, Networks & Communication Systems

2019 – 2023

Adventist University of Central Africa

Kigali, Rwanda

CERTIFICATIONS & LANGUAGES

Certifications AWS Certified Solutions Architect – Associate; AWS Certified Cloud Practitioner; Google Cybersecurity Professional Certificate

Languages French (native), English (full professional, EF SET C1), Spanish (elementary)